

Calories, macros & targets

How MiraFood calculates nutrition goals and scores meals

Plain-language product reference matching the Vitaway BP/Glucose formulas guide style. Rules from server NCE version 2026-07-clinical-v1. Stored targets always come from the server.

How MiraFood calculates nutrition goals and scores meals

This guide explains, in plain language, the formulas MiraFood uses when a patient onboards, when a coach confirms an assessment, and when the diary or dashboard scores the day.

All rules below come from the current server nutrition calculation engine (NCE_VERSION = 2026-07-clinical-v1). Mobile/web may show provisional previews during onboarding; stored clinical targets always come from the server.

1. What the patient enters

Field | Used for

Date of birth -> age | BMR age term; pediatric vs adult path

Sex | Mifflin / Schofield sex constants

Height (cm), weight (kg) | BMR, BMI, protein, water

Activity level | TDEE multiplier (adults / pregnancy / lactation)

Goal (+ optional pace) | Calorie surplus/deficit after coach confirmation

Meals per day | Consistency part of the health score

Preferences / allergies | Not used in calorie or macro math

Coach assessment (overrides)

When present, coach-verified values override patient basics:

- Verified age/sex/height/weight
- Pregnancy / trimester / number of babies / pre-pregnancy weight
- Lactation
- Conditions, fluid restriction
- Goal pace; optional coachAllowsProtectedWeightLoss

Target status

Status | Meaning

Unavailable | Missing age / height / weight / activity / goal

Provisional | Patient complete, or clinical flags still open

Confirmed | Coach confirmed assessment and no blocking flags

Rule: Goal deficits/surpluses apply only when status is confirmed. Until then, calorie target = maintenance TDEE.

2. BMI

$BMI = \text{weight_kg} / (\text{height_cm} / 100)^2$

Stored rounded to 1 decimal.

Flag: BMI over 40 -> bmi_over_40_coach_review (engine still uses actual weight in Mifflin-St Jeor).

3. BMR (basal metabolic rate)

Adults (age 18+) - Mifflin-St Jeor

$\text{base} = 10 \times \text{weight_kg} + 6.25 \times \text{height_cm} - 5 \times \text{age}$

Sex | BMR

Male | base + 5

Female | base - 161

Other / prefer not to say / unset | Average of male and female

Rounded to a whole number.

Pediatrics (under 18) - Schofield (weight-based)

Age | Boys | Girls

Under 3 | $59.512 \times W - 30.4$ | $58.317 \times W - 31.05$

3-9 | $22.706 \times W + 504.3$ | $20.315 \times W + 485.9$

10-17 | $17.686 \times W + 658.2$ | $13.384 \times W + 692.6$

Unknown sex -> average of boy and girl.

Pediatric TDEE = BMR (no activity multiplier until a validated pediatric reference is approved).

4. Activity multipliers

Activity level | Multiplier

Sedentary | 1.2

Lightly active | 1.375

Moderately active | 1.55

Very active | 1.725

Extremely active | 1.9

5. TDEE by population

Healthy adult

$TDEE = BMR \times activity_multiplier$

Pregnancy

- BMR from Mifflin using pre-pregnancy weight when available (else current weight + warning).

- Then:

$TDEE = BMR \times activity_multiplier + pregnancy_addition$

Case | Addition (kcal)

Trimester 1 (or missing) | 0

Singleton trimester 2 | +340

Singleton trimester 3 | +452

Twins, trimester 2 or 3 | +685

Lactation

$TDEE = BMR \times activity_multiplier + 500$

(BMR uses current weight.)

Pediatric

$TDEE = BMR$ (Schofield)

Rounded to a whole number.

6. Goal calorie adjustment

Applied only if coach assessment is confirmed:

Weight loss

Pace | Adjustment

Slow (default) | -500 kcal

Moderate | -625 kcal

Aggressive | -750 kcal

Muscle gain

Pace | Adjustment
Slow (default) | +300 kcal
Moderate | +400 kcal
Aggressive | +500 kcal

Maintain / improve diet quality

Adjustment = 0.

Protected populations

Pregnancy, lactation, and pediatric never get an automatic deficit unless the coach sets coachAllowsProtectedWeightLoss. Otherwise deficit is forced to 0 and flagged automatic_weight_loss_blocked.

Calorie target

calorieTarget = max(0, round(TDEE + adjustment))

7. Macro targets

Using calorie target C and weight W (kg):

Macro | Formula

Protein (g) | round($2.0 \times W$) if gain muscle; else round($1.6 \times W$)

Fat (g) | round($(0.28 \times C) / 9$) ~ 28% of calories

Carbs (g) | max(0, round($(C - 4 \times \text{protein} - 9 \times \text{fat}) / 4$))

Fiber (g/day) | 38 male; 25 female / other / unset

Energy check: $C \sim 4 \times \text{protein} + 4 \times \text{carbs} + 9 \times \text{fat}$ (carbs absorb rounding).

8. Water target

waterTargetMl = round(weight_kg x 35)

(~35 ml per kg). Flagged for clinical review if fluid restriction / kidney / heart disease; never overrides a prescribed restriction automatically.

9. Meal / portion nutrition

Case | Rule

Scale by weight | ratio = new_g / max(old_g, 1); multiply each nutrient

From per-100 g | nutrient(w) = nutrient_per_100g x (w / 100)

Meal totals | Sum of ingredient calories / P / C / F / fiber (+ micros when present)

What counts on the diary: only coach-approved meals for that calendar day.

10. Daily health score

Each component is 0-100.

Ratio score (calories, macros, nutrients)

r = actual / target

if r <= 1 -> score = 100 x r

if r > 1 -> score = 100 - 35 x (r - 1), then clamp

Weights

Component | Weight | Notes

Nutrient adequacy | 30% | Fiber always; iron / vitamin C / calcium / vitamin D / potassium when present

Macros | 25% | Average of protein, carbs, fat ratio scores

Calories | 20% | Consumed vs calorie target

Consistency | 15% | Approved meals today / meals-per-day goal

Variety | 10% | Distinct foods (target 5)

Missing micronutrients cannot look like perfect adequacy (coverage factor applied).

11. Simple decision trees

Which calorie path?

```
if age < 18 -> Schofield BMR; TDEE = BMR; protected
else if pregnant -> Mifflin (pre-pregnancy wt) x activity + trimester addition; protected
else if lactating -> Mifflin (current wt) x activity + 500; protected
else -> Mifflin x activity
```

Apply goal adjustment?

```
if assessment not confirmed -> adjustment = 0
else compute pace table
if adjustment < 0 and protected and not coach override -> adjustment = 0
calorieTarget = max(0, round(TDEE + adjustment))
```

Target status confirmed?

```
confirmed only if assessment confirmed
  AND no pediatric_review_required
  AND no missing pre-pregnancy weight / trimester
  AND sex confirmed for equation
  AND no individualized-target conditions blocking automation
else provisional
```

12. Client preview vs server (important)

| Mobile onboarding preview | Server NCE (stored)
Goal adjustment | Applied immediately (-500 / +300 defaults) | Only after coach confirm
Calorie floor | max(1200, TDEE + adj) | max(0, TDEE + adj)
Pregnancy / lactation / Schofield | Not in mobile util | Full NCE paths
Pace (slow/mod/aggressive) | Not in mobile util | Full NCE table

Treat onboarding numbers as estimates until coach confirmation.

13. Example (confirmed healthy adult male)

Input: 30 y, male, 180 cm, 80 kg, moderately active, maintain, confirmed.

```
BMR      = 10(80) + 6.25(180) - 5(30) + 5 = 1780
TDEE     = 1780 x 1.55 = 2759
C        = 2759
Protein  = round(1.6 x 80) = 128 g
Fat      = round(2759 x 0.28 / 9) = 86 g
Carbs    = round((2759 - 128x4 - 86x9) / 4) = 362 g
Fiber    = 38 g
Water    = round(80 x 35) = 2800 ml
BMI      = 24.7
```

(Matches nutrition-calculation-engine.test.ts.)

14. Where this lives in code

What | File

BMR / TDEE / macros / water / flags | server/src/modules/consumers/nutrition-calculation-engine.ts

Merge coach + patient -> calculate | server/src/modules/consumers/profile-targets.util.ts

Dashboard consumed + health score | server/src/modules/consumers/dashboard.util.ts

Micronutrient adequacy | server/src/modules/consumers/nutrient-score.util.ts
Portion scale / meal sum | server/src/modules/meals/nutrition.util.ts
Onboarding preview only | mobile/src/utils/nutrition.ts
Clinical policy overview | docs/clinical-nutrition-engine.md

Disclaimer

These formulas support educational coaching and logging. MiraFood is not a medical device and does not replace individualized clinical nutrition care. Protected populations and high-risk conditions require coach review before targets are treated as confirmed.

Internal product reference | From current app code | NCE 2026-07-clinical-v1